EEOB 546 R Assignment

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Remove objects you've created and load required libraries before running my code using this chunk:

```
rm(list = ls())
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Part I

1. Data Inspection

```
# load the dataframes
fang <- read_tsv ("https://raw.githubusercontent.com/EEOB-BioData/BCB5460_Spring2026/refs/heads/main/as
```

```
## Rows: 2782 Columns: 986
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
## chr (986): Sample_ID, JG_OTU, Group, abph1.20, abph1.22, ae1.3, ae1.4, ae1.5...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
snp <- read_tsv("https://raw.githubusercontent.com/EEOB-BioData/BCB5460_Spring2026/refs/heads/main/assi
```

```
## Rows: 983 Columns: 15
## -- Column specification --------------------------------------------------------
## Delimiter: "\t"
```

```
## chr (9): SNP_ID, Chromosome, Position, alt_pos, mult_positions, amplicon, cd...
## dbl (6): cdv_marker_id, Genaissance_daa_id, Sequenom_daa_id, count_amplicons...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(fang)
```

```
## # A tibble: 6 x 986
##   Sample_ID JG_OTU   Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6 ba1.9
##   <chr>     <chr>    <chr> <chr>    <chr>    <chr> <chr> <chr> <chr> <chr> <chr>
## 1 SL-15     T-aust-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   ?/?   G/G
## 2 SL-16     T-aust-2 TRIPS ?/?      ?/?      T/T   ?/?   T/T   C/C   A/G   G/G
## 3 SL-11     T-brav-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   ?/?   G/G   G/G
## 4 SL-12     T-brav-2 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   G/G   G/G
## 5 SL-18     T-cund   TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   ?/?   G/G
## 6 SL-2      T-dact-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   A/G   G/G
## # i 975 more variables: bt2.5 <chr>, bt2.7 <chr>, bt2.8 <chr>, Fea2.1 <chr>,
## #   Fea2.5 <chr>, id1.3 <chr>, lg2.11 <chr>, lg2.2 <chr>, pbf1.1 <chr>,
## #   pbf1.2 <chr>, pbf1.3 <chr>, pbf1.5 <chr>, pbf1.6 <chr>, pbf1.7 <chr>,
## #   pbf1.8 <chr>, PZA00003.11 <chr>, PZA00004.2 <chr>, PZA00005.8 <chr>,
## #   PZA00005.9 <chr>, PZA00006.13 <chr>, PZA00006.14 <chr>, PZA00008.1 <chr>,
## #   PZA00010.5 <chr>, PZA00013.10 <chr>, PZA00013.11 <chr>, PZA00013.9 <chr>,
## #   PZA00015.4 <chr>, PZA00017.1 <chr>, PZA00018.5 <chr>, ...
```

```r
ncol(fang)
```

```
## [1] 986
```

```r
nrow(fang)
```

```
## [1] 2782
```

```r
dim(fang)
```

```
## [1] 2782  986
```

```r
head(snp)
```

```
## # A tibble: 6 x 15
##   SNP_ID   cdv_marker_id Chromosome Position  alt_pos mult_positions amplicon
##   <chr>            <dbl> <chr>      <chr>     <chr>   <chr>          <chr>
## 1 abph1.20          5976 2          27403404  <NA>    <NA>           abph1
## 2 abph1.22          5978 2          27403892  <NA>    <NA>           abph1
## 3 ae1.3             6605 5          167889790 <NA>    <NA>           ae1
## 4 ae1.4             6606 5          167889682 <NA>    <NA>           ae1
## 5 ae1.5             6607 5          167889821 <NA>    <NA>           ae1
## 6 an1.4             5982 1          240498509 <NA>    <NA>           an1
## # i 8 more variables: cdv_map_feature.name <chr>, gene <chr>,
## #   `candidate/random` <chr>, Genaissance_daa_id <dbl>, Sequenom_daa_id <dbl>,
## #   count_amplicons <dbl>, count_cmf <dbl>, count_gene <dbl>
```

```r
ncol(snp)
```

```
## [1] 15
```

```r
nrow(snp)
```

```
## [1] 983
```

```r
dim(snp)
```

```
## [1] 983  15
```

```r
colnames(snp)
```

```
##  [1] "SNP_ID"               "cdv_marker_id"        "Chromosome"
##  [4] "Position"             "alt_pos"              "mult_positions"
##  [7] "amplicon"             "cdv_map_feature.name" "gene"
## [10] "candidate/random"     "Genaissance_daa_id"   "Sequenom_daa_id"
## [13] "count_amplicons"      "count_cmf"            "count_gene"
```

fang *We have 986 columns, and 2782 rows. Columns include Sample_ID, individual genotypes, and metadata.* **snp** *We have 15 columns, and 983 rows. These rows are the SNPs. The columns includes SNP Id, chromosome and nucleotide position*

---

2. Data Processing

Transposing and merging

```r
# Transpose fang_et_el data
fang_t <- as.data.frame(fang)
fang_t$SNP_ID <- rownames(fang_t)

# make the genotypes data a "long format" for easier manipulation.
fang_long <- fang_t %>%
  pivot_longer(
    cols = 4:ncol(fang_t),
    names_to = "SNP_ID",
    values_to = "Genotype"
  )

head(fang_long)
```

```
## # A tibble: 6 x 5
##   Sample_ID JG_OTU   Group SNP_ID   Genotype
##   <chr>     <chr>    <chr> <chr>    <chr>
## 1 SL-15     T-aust-1 TRIPS abph1.20 ?/?
```

```
## 2 SL-15     T-aust-1 TRIPS abph1.22 ?/?
## 3 SL-15     T-aust-1 TRIPS ae1.3    T/T
## 4 SL-15     T-aust-1 TRIPS ae1.4    G/G
## 5 SL-15     T-aust-1 TRIPS ae1.5    T/T
## 6 SL-15     T-aust-1 TRIPS an1.4    C/C
```

```r
dim(fang_long)
```

```
## [1] 2737488       5
```

```r
# join genotypes data with the snp positions by SNP_ID
merged_data <- inner_join(fang_long, snp, by = "SNP_ID")
dim(merged_data)
```

```
## [1] 2734706      19
```

```r
head(merged_data)
```

```
## # A tibble: 6 x 19
##   Sample_ID JG_OTU   Group SNP_ID   Genotype cdv_marker_id Chromosome Position
##   <chr>     <chr>    <chr> <chr>    <chr>            <dbl> <chr>      <chr>
## 1 SL-15     T-aust-1 TRIPS abph1.20 ?/?               5976 2          27403404
## 2 SL-15     T-aust-1 TRIPS abph1.22 ?/?               5978 2          27403892
## 3 SL-15     T-aust-1 TRIPS ae1.3    T/T               6605 5          167889790
## 4 SL-15     T-aust-1 TRIPS ae1.4    G/G               6606 5          167889682
## 5 SL-15     T-aust-1 TRIPS ae1.5    T/T               6607 5          167889821
## 6 SL-15     T-aust-1 TRIPS an1.4    C/C               5982 1          240498509
## # i 11 more variables: alt_pos <chr>, mult_positions <chr>, amplicon <chr>,
## #   cdv_map_feature.name <chr>, gene <chr>, `candidate/random` <chr>,
## #   Genaissance_daa_id <dbl>, Sequenom_daa_id <dbl>, count_amplicons <dbl>,
## #   count_cmf <dbl>, count_gene <dbl>
```

```r
# Convert back to "wide format" with `SNP_ID`, `Chromosome`, `Position`, and sample genotypes as column
merged_wide <- merged_data %>%
  select(SNP_ID, Chromosome, Position, Sample_ID, Genotype) %>%
  pivot_wider(
    names_from = Sample_ID,
    values_from = Genotype
  )

dim(merged_wide)
```

```
## [1]  983 2785
```

```r
head(merged_wide)
```

```
## # A tibble: 6 x 2,785
##   SNP_ID   Chromosome Position  `SL-15` `SL-16` `SL-11` `SL-12` `SL-18` `SL-2`
##   <chr>    <chr>      <chr>     <chr>   <chr>   <chr>   <chr>   <chr>   <chr>
## 1 abph1.20 2          27403404  ?/?     ?/?     ?/?     ?/?     ?/?     ?/?
```

```
## 2 abph1.22 2          27403892  ?/?     ?/?     ?/?     ?/?     ?/?     ?/?
## 3 ae1.3    5          167889790 T/T     T/T     T/T     T/T     T/T     T/T
## 4 ae1.4    5          167889682 G/G     ?/?     G/G     G/G     G/G     G/G
## 5 ae1.5    5          167889821 T/T     T/T     T/T     T/T     T/T     T/T
## 6 an1.4    1          240498509 C/C     C/C     ?/?     C/C     C/C     C/C
## # i 2,776 more variables: ‘SL-4‘ <chr>, ‘SL-6‘ <chr>, ‘SL-7‘ <chr>,
## #   ‘SL-5‘ <chr>, ‘SL-3‘ <chr>, ‘SL-8‘ <chr>, ‘SL-24‘ <chr>, ‘SL-22‘ <chr>,
## #   ‘SL-14‘ <chr>, ‘SL-23‘ <chr>, ‘SL-20‘ <chr>, ‘SL-19‘ <chr>, ‘SL-17‘ <chr>,
## #   ‘SL-21‘ <chr>, ‘SL-9‘ <chr>, ‘SL-10‘ <chr>, ZDP_1280a <chr>,
## #   ZDP_1312a <chr>, ZDP_1454b <chr>, ‘JVLJ-690-1‘ <chr>, ‘JVLJ-690-19‘ <chr>,
## #   ‘JVLJ-690-3‘ <chr>, ‘JVLJ-690-5‘ <chr>, ‘JVLJ-690-7‘ <chr>,
## #   ‘JVLJ-690-9‘ <chr>, ‘JVLJ-690-11‘ <chr>, ‘JVLJ-690-13‘ <chr>, ...
```

Split maize and teosinte groups

```r
# Split datasets into "maize" and "teosinte" groups
maize_samples <- fang_t %>%
  filter(Group %in% c("ZMMIL","ZMMLR","ZMMMR")) %>%
  pull(Sample_ID)

teosinte_samples <- fang_t %>%
  filter(Group %in% c("ZMPBA","ZMPIL","ZMPJA")) %>%
  pull(Sample_ID)

# subset
maize <- merged_wide %>%
  select(SNP_ID, Chromosome, Position, all_of(maize_samples))

teosinte <- merged_wide %>%
  select(SNP_ID, Chromosome, Position, all_of(teosinte_samples))
```

Geneerate chromosomes files

```r
# Generate 40 files
write_chr_files <- function(data, group_name){

lapply(1:10, function(chr){

  chr_data <- data %>%
  filter(Chromosome == chr)

# Increasing position
inc <- chr_data %>%
  arrange(Position) %>%
  mutate(across(4:ncol(.), ~replace_na(., "?")))

write.table(
  inc,
  paste0("output_files/", group_name, "_chr", chr, "_increasing.txt"),
  sep="\t",
```

```r
  row.names=FALSE,
  quote=FALSE
  )

# Decreasing position
dec <- chr_data %>%
  arrange(desc(Position)) %>%
  mutate(across(4:ncol(.), ~replace_na(., "-")))

write.table(
  dec,
  paste0("output_files/", group_name, "_chr", chr, "_decreasing.txt"),
  sep="\t",
  row.names=FALSE,
  quote=FALSE
  )
  })
}
colnames(maize) [1:3]
```

```
## [1] "SNP_ID"     "Chromosome" "Position"
```

```r
# Inspecting produced files;
# Calling out a random file to check them out
m_chr8_d <- read.table("output_files/maize_chr8_decreasing.txt", header = T)
colnames(m_chr8_d) [1:3]
```

```
## [1] "SNP_ID"     "Chromosome" "Position"
```

Note that these files were saved in my folder called "output_files", so if you want to regenerate the files into your working directory, replace the "output_files" in the code with whatever the folder you would like to save them in your current working directory. Or create a folder called "output_files" first. Thanks.

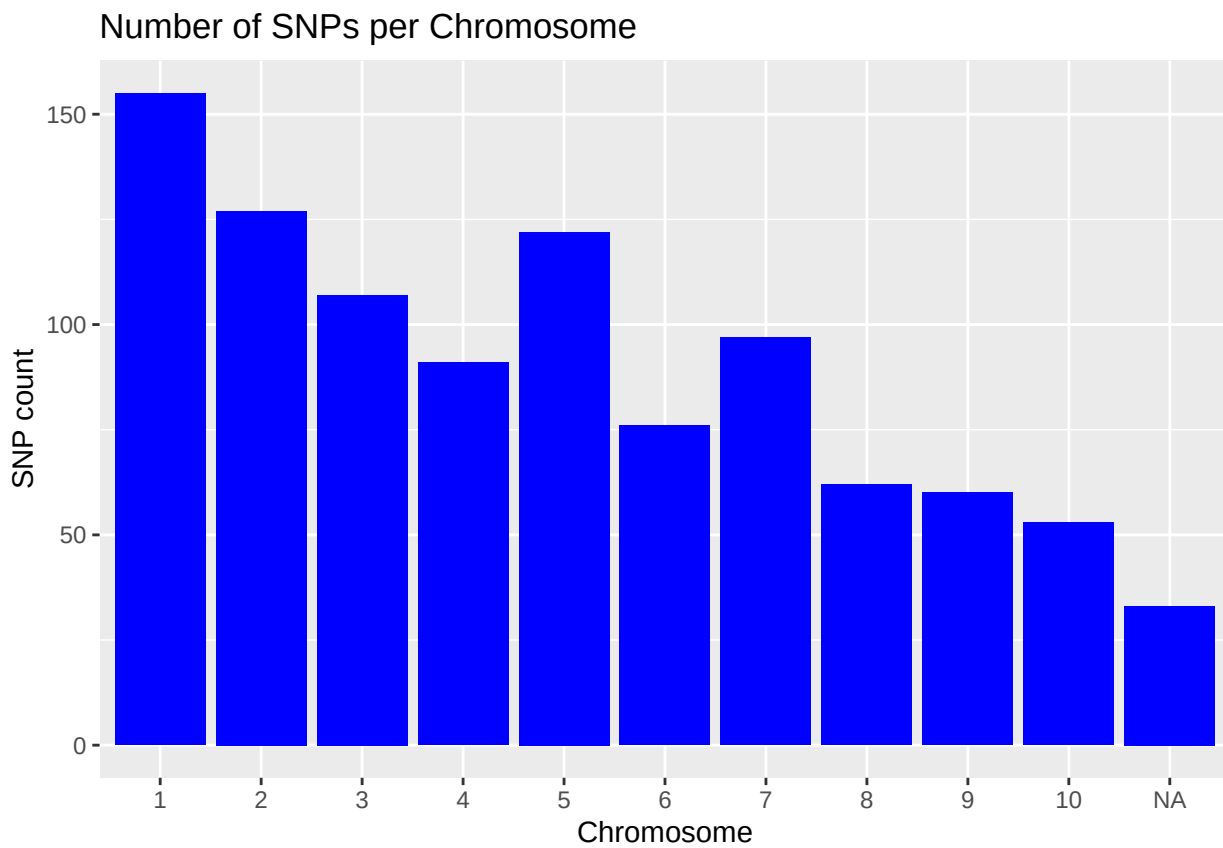
Part II

Visualization

SNPs per chromosome

```r
# SNPs per chromosome
merged_wide %>%
  count(Chromosome) %>%
  mutate(Chromosome = factor(Chromosome, levels = sort(unique(as.numeric(Chromosome))))) %>%
  ggplot(aes(x = Chromosome, y = n)) +
  geom_bar(stat = "identity", fill = "blue") +
  labs(
    title = "Number of SNPs per Chromosome",
    x = "Chromosome",
    y = "SNP count"
  )
```

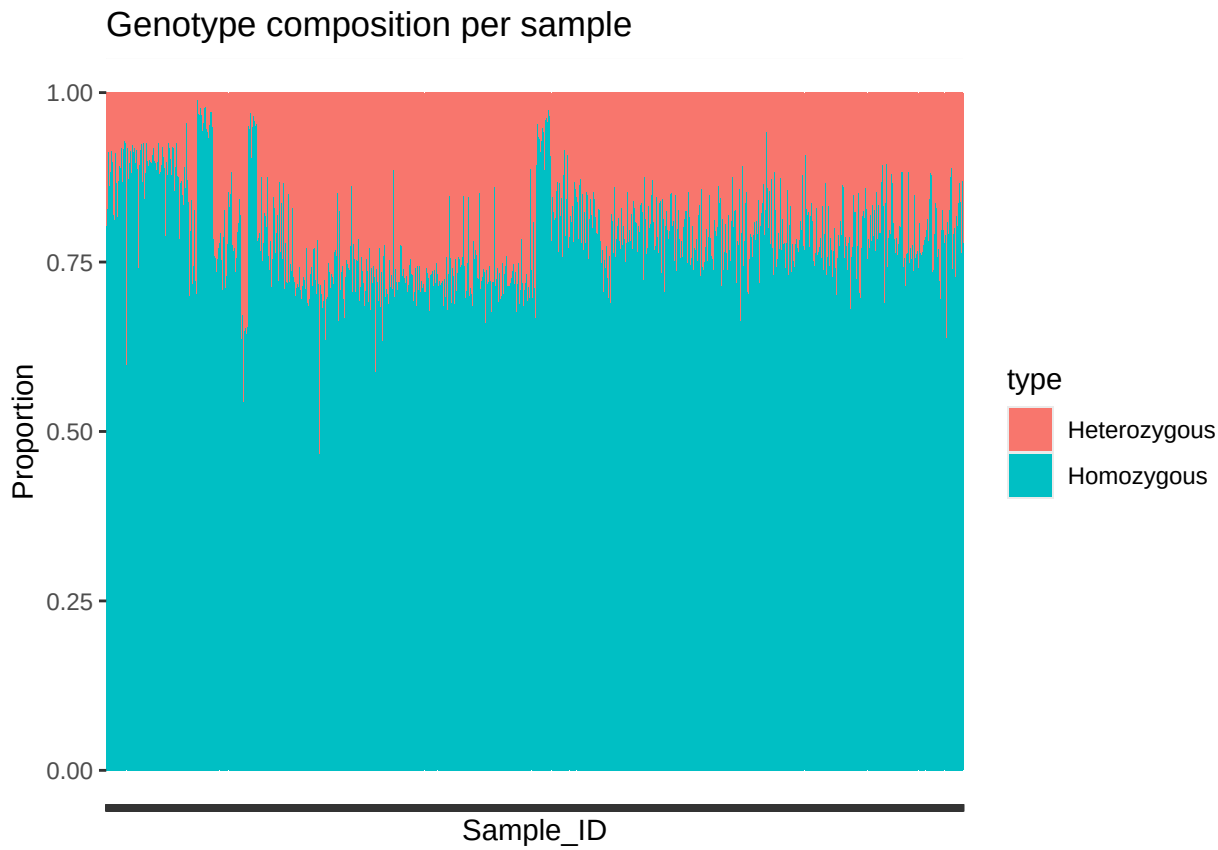
```
## Warning: There was 1 warning in ‘mutate()‘.
## i In argument: ‘Chromosome = factor(Chromosome, levels =
##   sort(unique(as.numeric(Chromosome))))‘.
## Caused by warning in ‘unique()‘:
## ! NAs introduced by coercion
```



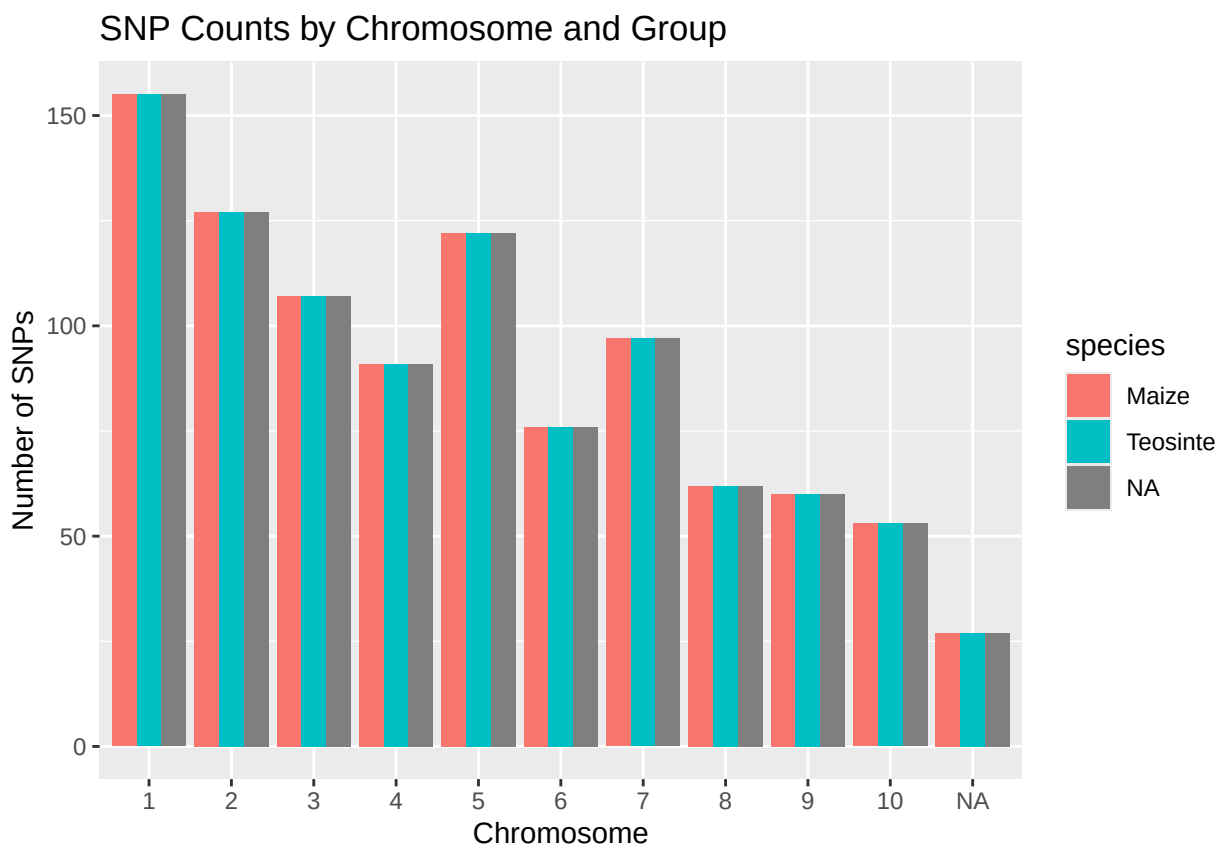
Missing data and amount of heterozygosity

```r
# Missing data and amount of heterozygosity
geno_long <- merged_data %>%
  mutate(
    type = case_when(
      Genotype %in% c("A/A","C/C","G/G","T/T") ~ "Homozygous",
      is.na(Genotype) ~ "Missing",
      TRUE ~ "Heterozygous"
    )
  )


ggplot(geno_long, aes(x=Sample_ID, fill=type)) +
  geom_bar(position="fill") +
  labs(
    y="Proportion",
    title="Genotype composition per sample"
```

```
  ) +
  theme(axis.text.x = element_blank())
```



Create my own visualization ### SNP counts by chromosome in maize vs teosinte

```
# SNP counts by chromosome in maize vs teosinte
geno_long <- merged_data %>%
  mutate(
    species = case_when(
      Group %in% c("ZMMIL","ZMMLR","ZMMMR") ~ "Maize",
      Group %in% c("ZMPBA","ZMPIL","ZMPJA") ~ "Teosinte"
    )
  )

geno_long %>%
  distinct(SNP_ID, Chromosome, species) %>%
  count(Chromosome, species) %>%
  mutate(Chromosome = factor(Chromosome,
                             levels = sort(unique(as.numeric(Chromosome))))) %>%
  ggplot(aes(x = Chromosome, y = n, fill = species)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "SNP Counts by Chromosome and Group",
    x = "Chromosome",
    y = "Number of SNPs"
  )
```

```
## Warning: There was 1 warning in ‘mutate()‘.
## i In argument: ‘Chromosome = factor(Chromosome, levels =
##   sort(unique(as.numeric(Chromosome))))‘.
## Caused by warning in ‘unique()‘:
## ! NAs introduced by coercion
```